

Alexandre Branco

Ph.D. Candidate in Astronomy, University of Porto, Portugal

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Education

Ph.D. in Astronomy

Jan 2025 - Sep 2028 (expected)

University of Porto, Portugal

- Thesis: *Towards a Comprehensive Understanding of the Atmospheric Composition of Small Exoplanets*. Advisors: Dr. Clara Sousa-Silva (Bard College and IA), Dr. Olivier Demangeon (IA) & Dr. Pedro Machado (University of Lisbon and IA).

M.Sc. in Physics – Specialization in Astrophysics and Cosmology

Sep 2022 - Jan 2025

University of Lisbon, Portugal

Final Grade: 19/20 (A¹)

- Thesis: *From Venus to Ultra-Hot Jupiters: Characterizing Planetary Atmospheres with High-Resolution Transmission Spectroscopy*. Advisors: Dr. Pedro Machado, Dr. Olivier Demangeon & Dr. Tomás Azevedo Silva. Grade: 19/20.

B.Sc. in Physics – Specialization in Astronomy and Astrophysics

Sep 2019 - Jul 2022

University of Lisbon, Portugal

Final Grade: 16/20 (B¹)

- Research Project²: *High-Resolution Transmission Spectroscopy of WASP-76 b*. Advisor: Dr. Pedro Machado. Grade: 18/20.

¹Grade according to the ECTS grading scale: A - Excellent, B - Very Good, C - Good, D - Satisfactory, E - Sufficient;

²At the University of Lisbon there is no bachelor's thesis in physics, but a semester project (6 ECTS).

Research Experience

Ph.D. Candidate

Jan 2025 - Sep 2028 (expected)

Institute of Astrophysics and Space Sciences, Porto, Portugal

- Combining quantum chemistry, photochemical modelling, and atmospheric retrieval codes to investigate ambiguities in exoplanet atmospheric studies. Research includes the impact of molecular cross sections and chemical kinetics on planetary habitability assessments, and on the interpretation of observations of rocky worlds and sub-Neptunes.

M.Sc. Student

Sep 2023 - Sep 2024

Institute of Astrophysics and Space Sciences, Lisbon, Portugal

- Analysed DST/FIRS observations of the Venus solar transit of 2012 to detect key chemical species in its transmission spectrum. Used spectroscopic databases and the petitRADTRANS code to highlight potential spectral observables for Venus-like exoplanet atmospheres. Also analysed VLT/ESPRESSO transit observations of the ultra-hot Jupiter WASP-76 b. Identified and characterized atomic absorption features, revealing the planet's atmospheric dynamics.

Visiting Scholar

Oct 2023 - Sep 2024

Blue Marble Space Institute of Science (remote)

- Used computational chemistry to model complex chemical reaction networks, involving thousands of molecules. I aimed to simulate the reaction of glucose with hydrogen sulfide to study the synthesis of complex sulfur-bearing compounds from relatively simple organic molecules, helping to predict key processes in the origin of life. Advisor: Dr. Jim Cleaves (Carnegie Institution for Science and Howard University).

Research Associate

Jun - Sep 2022

Blue Marble Space Institute of Science (remote)

- Completed the Young Scientist Program, including science communication and ethics modules. Conducted a scientific literature analysis on surface reflectance biosignatures linked to the pigmentation of extremophiles. Advisor: Dr. Rafael Loureiro (Winston-Salem State University).

Laboratory of Instrumentation and Experimental Particle Physics, Lisbon, Portugal

- Worked with Galactic Cosmic Ray models and Monte Carlo simulations to study the interactions between highly energetic particles and semiconductor materials. I aimed to address the feasibility of SRAM-based devices as radiation monitors at the orbit and surface of Mars. Advisors: Dr. Marco Pinto (ESA) and Dr. Luísa Arruda (LIP).

Publications

First-Authored

- **Branco, A.**; Sousa-Silva, C.; Broussard, W.; Schwieterman, E.; *et al.*, "Assessing the Impact of Varying HSO Cross Sections on Photochemical Models: Implications for the Spectral Characterization of Terrestrial Exoplanets", *The Astrophysical Journal*, vol. 1007, 2026.  DOI: 10.3847/1538-4357/ae89b5.
- **Branco, A.**; Machado, P.; Demangeon, O.; Azevedo Silva, T.; *et al.*, "Transmission Spectroscopy Along the Transit of Venus: A Proxy for Exoplanets Atmospheric Characterization", *Atmosphere MDPI*, vol. 15, 2024.  DOI: 10.3390/atmos15121431.

Co-Authored

- Dressing, C.; Adams, D.; Alecian, E.; Anand, G.; *et al.*, "Early Exploration of the Scientific Discovery Space for the Habitable Worlds Observatory", *Space Science Reviews*, submitted.  arXiv: 2608.11294.

Talks & Conference Contributions

- **XXXVI Encontro Nacional de Astronomia e Astrofísica**, Porto, Portugal, Sep 09-12, 2026 (contributed talk)
Branco, A., Sousa-Silva, C., Broussard, W., *et al.*, "The Fear of Missing Out in Atmospheric Studies: Exploring the Role of Molecular Opacities in Interpreting Exoplanet Spectra".
- **HoRS3S Conference**, Granada, Spain, Jul 06-10, 2026 (contributed talk)
Branco, A., Sousa-Silva, C., Broussard, W., *et al.*, "The Fear of Missing Out in Atmospheric Studies: Exploring the Role of Molecular Opacities in Interpreting Exoplanet Spectra".
- **Exoplanets 6**, Porto, Portugal, Jun 29-Jul 03, 2026 (poster)
Branco, A., Sousa-Silva, C., Broussard, W., *et al.*, "The Fear of Missing Out in Atmospheric Studies: Exploring the Role of Molecular Opacities in Interpreting Exoplanet Spectra".
- **Cookie Seminar @IA**, Porto, Portugal, Jun 08, 2026
Branco, A. & Mota, A., "Scientific Potential of the Future Flagship Telescopes HWO and LIFE".
- **Rocky Worlds 4**, Groningen, the Netherlands, Jan 19-23, 2026 (poster)
Branco, A., Sousa-Silva, C., Broussard, W., *et al.*, "Assessing the Impact of Varying HSO and HNO Cross-Sections on Photochemical Models: Implications for Spectral Characterization of Rocky Worlds".
- **EANA 2025 Conference**, Lisbon, Portugal, Oct 21-24, 2025 (contributed talk)
Branco, A., Sousa-Silva, C., Broussard, W., *et al.*, "Exploring the Impact of HSO and HNO Cross-Section Variability on Photochemical Modelling and its Implications for the Spectral Characterization of Terrestrial Exoplanets".
- **EPSC-DPS Joint Meeting 2025**, Helsinki, Finland, Sep 07-12, 2025 (contributed talk)
Branco, A., Machado, P., Demangeon, O., *et al.*, "High-Resolution Transmission Spectroscopy of Venus: A Proxy for Atmospheric Characterization of Earth-Sized Exoplanets".
- **EPSC-DPS Joint Meeting 2025**, Helsinki, Finland, Sep 07-12, 2025 (poster)
Branco, A., Sousa-Silva, C., Broussard, W., *et al.*, "Assessing the Impact of Varying HSO and HNO Cross-Sections on Photochemical Models: Implications for the Spectral Characterization of Terrestrial Exoplanets".
- **Europlanet Webinar: Transit of Venus 2012 – Past and Present**, online, Dec 11, 2024
Branco, A., Machado, P., Demangeon, O., *et al.*, "Transmission Spectroscopy Along the Transit of Venus: A Proxy for Exoplanets Atmospheric Characterization".

Meetings, Workshops & Advanced Schools

- **Summer School Alpbach 2026**, Alpbach, Austria, Jul 14-23, 2026.
 - * Science Team Lead for ICARUS mission – *Investigating the Chronology of Stellar Activity and Radiation Using UV Spectra*.
 - * Mission concept awarded *Best Science Case Prize*.
 - * Mission concept selected for the Post Alpbach Summer School Event.
- **Characterizing Habitable Exoplanets with Interdisciplinary Expertise (CHEWIE) Workshop**, Leiden, the Netherlands, Apr 28-May 01, 2026.
- **XXI International Astrobiology Summer School "Josep Comas i Solà"**, Santander, Spain, Jul 07-11, 2025.
- **Ariel Spring Consortium Meeting 2025**, Leiden, the Netherlands, Apr 09-11, 2025.
- **Ariel Fall Consortium Meeting 2024**, Lisbon, Portugal, Oct 28-30, 2024.
- **III Course of Environmental Sampling**, University of Lisbon, Portugal, Sep 04-08, 2023.

Research Collaborations & Memberships

- **Member of the LIFE Mission Science Team.**
- **Member of the Ariel Mission Consortium.**
- **Member of the Ariel Chemistry Working Group.**
- **Member of the Ariel Molecular Spectroscopy Working Group.**
- **Member of the ELT Working Group on Advanced Signal Processing of Exoplanet Spectroscopy (ASPEX).**
- **Served as acting Steering Committee member of the HWO Living Worlds Working Group**, Apr-Jul 2025.
- **Member of NASA project on Experimental Constraints for Improving Terrestrial Exoplanet Photochemical Models (ExCITE-PM)**, NASA Exoplanets Research Program - PI: S. Ranjan & E. Schwieterman.

Organisational Roles

- **Local Organising Committee member for Exoplanets 6 Conference**, Porto, Portugal, Jun 29-Jul 03, 2026.
- **Co-Host of Hack-4-Sages – Digital Twins in Astrobiology Hackathon**, Porto, Portugal, Mar 09-13, 2026.
- **Host of Breakout Session *Detection and Interpretation of Prebiosignature Gases at Rocky Worlds 4***, Groningen, the Netherlands, Jan 19-23, 2026.
- **Local Organising Committee member for EANA 2025 Conference**, Lisbon, Portugal, Oct 21-24, 2025.
- **Local Organising Committee member for Ariel Fall Consortium Meeting 2024**, Lisbon, Portugal, Oct 28-30, 2024.

Science Communication & Supervision

- **Invited scientist at *Pint of Science Porto***, May 20, 2026, Porto, Portugal.
- **Invited scientist at Hackathon "Exoplanetas em Bytes"**, Porto, Portugal, May 13, 2026.
- **Member of the AstroBioLusitanos Science Communication Team**, Oct 2025-ongoing.
Participating in science communication and outreach activities in Portugal.
- **Invited scientist at Hackathon "Exoplanetas em Bytes"**, Lisbon, Portugal, Oct 04, 2024.
- **Co-supervisor of Margarida Graça, B.Sc. research project**, University of Lisbon, Dec 2023-Jun 2024.
Title: *WASP-76 b Atmosphere's Characterization Using Transmission Spectroscopy*.

Awards & Grants

- **ESA Academy Scholarship**, awarded to 5 participants to attend the XXI International Astrobiology Summer School "Josep Comas i Solà", 2025.
- **Ph.D. Research Fellowship**, 4-year fellowship co-funded by the Heising-Simons Foundation and Bard College, 2024.

Other Skills

Programming	Python (B.Sc.–Ph.D. coursework and research); C++ (B.Sc. coursework)
Languages	Portuguese: Native; English: CEFR Level C1